

Structure of Atom

SHORT NOTES

Charge to Mass Ratio of Fundamental Particles

Proton (m_p)/anode rays	Neutron (m_n)	Electron (m_e)/cathode rays
mass = 1.67×10^{-27} kg	mass = 1.67×10^{-27} kg	mass = 9.1×10^{-31} kg
mass = 1.67×10^{-24} g	mass = 1.67×10^{-24} g	mass = 9.1×10^{-28} g
mass = 1.00727 amu	mass = 1.00867 amu	mass = 0.000549 amu
e/m value is dependent on the nature of gas taken in discharge tube.		e/m of electron is found to be independent of nature of gas & electrode used.

Electromagnetic Spectrum

- ❖ RW $\rightarrow$ MW $\rightarrow$ IR $\rightarrow$ Visible $\rightarrow$ UV $\rightarrow$ X-rays $\rightarrow$ CR (Radiowaves $\rightarrow$ Microwaves $\rightarrow$ Infrared rays $\rightarrow$ Visible rays $\rightarrow$ Ultraviolet rays $\rightarrow$ X-rays $\rightarrow$ Cosmic rays)
- ❖ Wavelength decreases $\rightarrow$
- ❖ Frequency increases $\rightarrow$
- ❖ $c = \nu\lambda$ $\lambda = \frac{c}{\nu}$ $\frac{1}{\nu} = \frac{\lambda}{c} = \frac{v}{c}$
- $T = \frac{1}{\nu}$ $E = \frac{hc}{\lambda} = h\nu$ $h = 6.626 \times 10^{-34}$ Js
- $E(\text{ev}) = \frac{12400}{\lambda(\text{\AA})}$
- ❖ Total amount of energy transmitted $E = nh\nu = \frac{nhc}{\lambda}$

Bohr's Atomic Model

Theory based on quantum theory of radiation and the classical laws of physics

- ❖ **Radius:** $r = 0.529 \times \frac{n^2}{Z} \text{\AA}$
- ❖ **Velocity:** $v = 2.188 \times 10^6 \frac{Z}{n} \text{ms}^{-1}$
- ❖ Energy (KE + PE) = Total energy = $-13.6 \times \frac{Z^2}{n^2} \text{eV/atom}$
- ❖ $TE = -\frac{KZe^2}{2r}$, $PE = \frac{-KZe^2}{r}$, $KE = \frac{KZe^2}{2r}$
 $PE = -2KE$, $KE = -TE$, $PE = 2TE$
- ❖ Revolutions per sec = $\frac{v}{2\pi r} \propto \frac{Z^2}{n^3}$
- ❖ Time for one revolution = $\frac{2\pi r}{v} \propto \frac{n^3}{Z^2}$
- ❖ Energy difference between n_1 and n_2 energy level is given by:

$$\Delta E = E_{n_2} - E_{n_1} = 13.6Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \frac{\text{eV}}{\text{atom}} = IE \times \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

where, IE = ionization energy of single electron species.

$$\text{❖ Ionization energy} = E_{\infty} - E_{G.S.} = 0 - E_{G.S.}$$

$E_{G.S.}$ = Energy of electron in ground state

$$E = -2.18 \times 10^{-11} \frac{Z^2}{n^2} \text{ erg per atom}$$

$$= -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J per atom}$$

$$= -13.6 \frac{Z^2}{n^2} \text{ eV/atom}$$

$$1 \text{ eV} = 3.83 \times 10^{-23} \text{ kcal}$$

$$1 \text{ eV} = 1.602 \times 10^{-12} \text{ erg}$$

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

$$E = -313.6 \frac{Z^2}{n^2} \text{ kcal/mole (1 cal = 4.18 J)}$$

$$V = \frac{2\pi kZe^2}{nh} \quad V \propto \frac{Z}{n}$$

$$r = \frac{n^2 h^2}{4\pi^2 m k Z e^2} \quad r \propto \frac{n^2}{Z}$$

$$E = \frac{-2\pi^2 m Z^2 e^4 k^2}{n^2 h^2} \quad E \propto \frac{Z^2}{n^2}$$

Hydrogen Spectrum

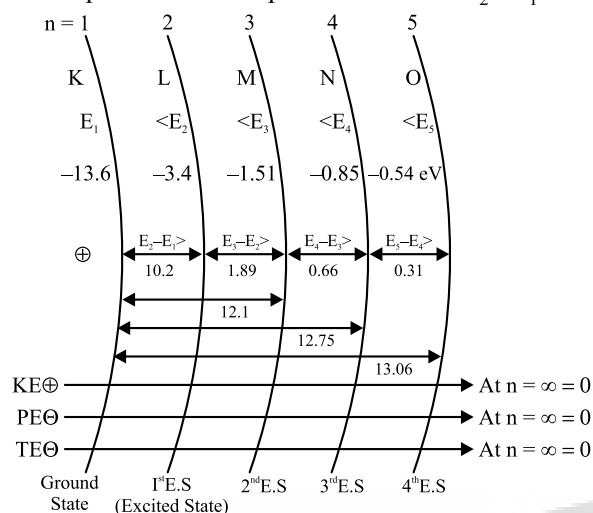
❖ **Rydberg's Equation:**

$$\frac{1}{\lambda} = \bar{\nu} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \times Z^2$$

$R_H \cong 109700 \text{ cm}^{-1}$ = Rydberg constant

- ❖ For first line of a series, $n_2 = n_1 + 1$
- ❖ Limiting spectral line (series limit) means $n_2 = \infty$
- ❖ H_{α} line means $n_2 = n + 1$; also known as line of longest λ , shortest ν , least E
- ❖ Similarly, H_{β} line means $n_2 = n_1 + 2$
- ❖ When electron de-excite from higher energy level (n) to ground state in atomic sample, then number of spectral lines observed in the spectrum = $\frac{n(n-1)}{2}$
- ❖ When electrons de-excite from higher energy level (n_2) to lower energy level (n_1) in atomic sample, then number of spectral line observed in the spectrum = $\frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$

- ❖ No. of spectral lines in a particular series = $n_2 - n_1$



De-Broglie Hypothesis

- ❖ All material particles possess wave character as well as particle character.

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

- ❖ The circumference of the n^{th} orbit is equal to n times of wavelength of electron i.e., $2\pi r_n = n\lambda$

Number of waves = n = principal quantum number

- ❖ Wavelength of electron (λ) $\cong \sqrt{\frac{150}{V(\text{volts})}} \text{ Å}$ for electron

- ❖ $\lambda = \frac{h}{\sqrt{2mKE}}$ For a particle $\lambda = \frac{0.101}{\sqrt{V}} \text{ Å}$

Heisenberg Uncertainty Principle

- ❖ According to this principle, "it is impossible to measure simultaneously the position and momentum of a microscopic particle with absolute accuracy".

If one of them is measured with greater accuracy, the other becomes less accurate.

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi} \quad \text{or} \quad (\Delta x)(\Delta v) \geq \frac{h}{4\pi m}$$

where Δx = Uncertainty in position
 Δp = Uncertainty in momentum
 Δv = Uncertainty in velocity
 m = Mass of microscopic particle

- ❖ Heisenberg replaced the concept of orbit by that of orbital.

Rules for Filling of Orbitals

- ❖ **Aufbau principle:** The electrons are filled up in increasing order of the energy in subshells.

$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^6 6s^2 4f^{14} 5d^{10} 6p^6 7s^2 5f^{14} 6d^{10}$

- ❖ **($n + l$) rule:** The subshell with lowest ($n + l$) value is filled up first, but when two or more subshells have same ($n + l$) value, then the subshell with lowest value of n is filled up first.

- ❖ **Pauli exclusion principle:** Pauli stated that no two electrons in an atom can have same values of all four quantum numbers.

- ❖ **Hund's rule of maximum multiplicity:** Electrons are distributed among the orbitals of subshell in such a way as to give maximum number of unpaired electrons with parallel spins.

Nodes

Total nodes = $n - 1$

Angular nodes = ℓ

Radial node = $n - \ell - 1$